

Thoughts on Important Challenges in Network Research

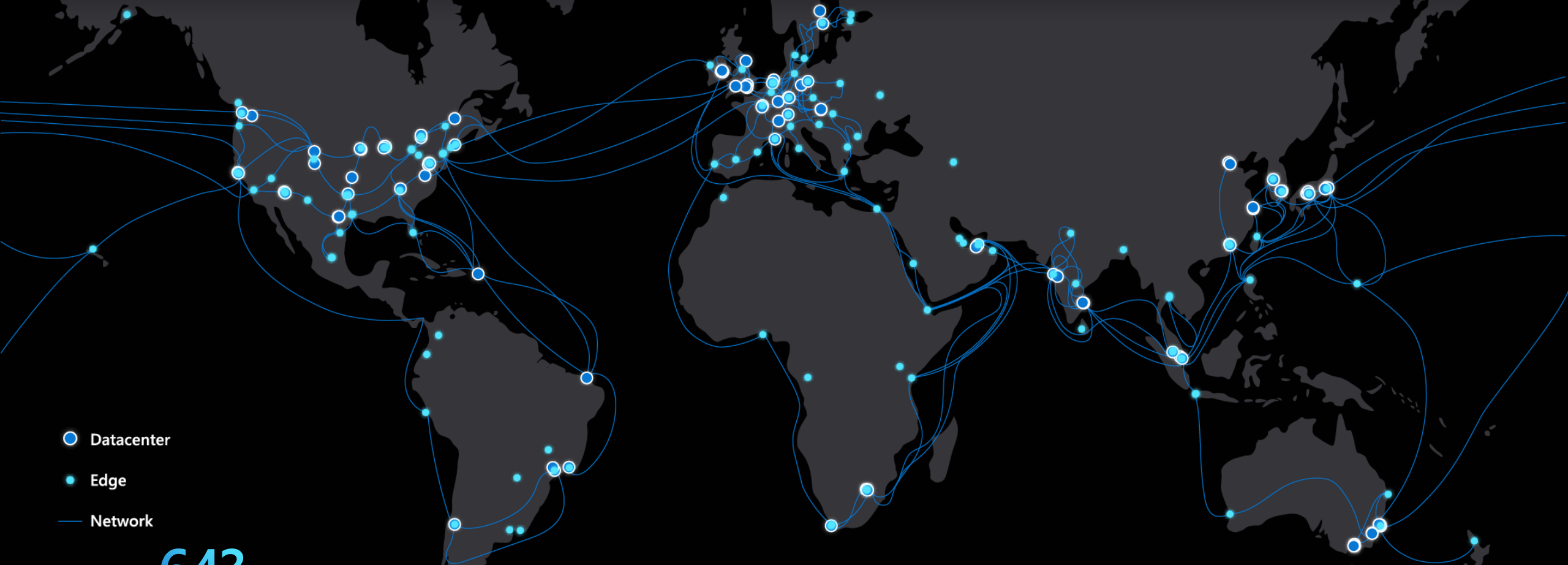
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(not representing Microsoft)

Microsoft Global Network



- Datacenter
- Edge
- Network

~~61~~ **64?**
Azure regions

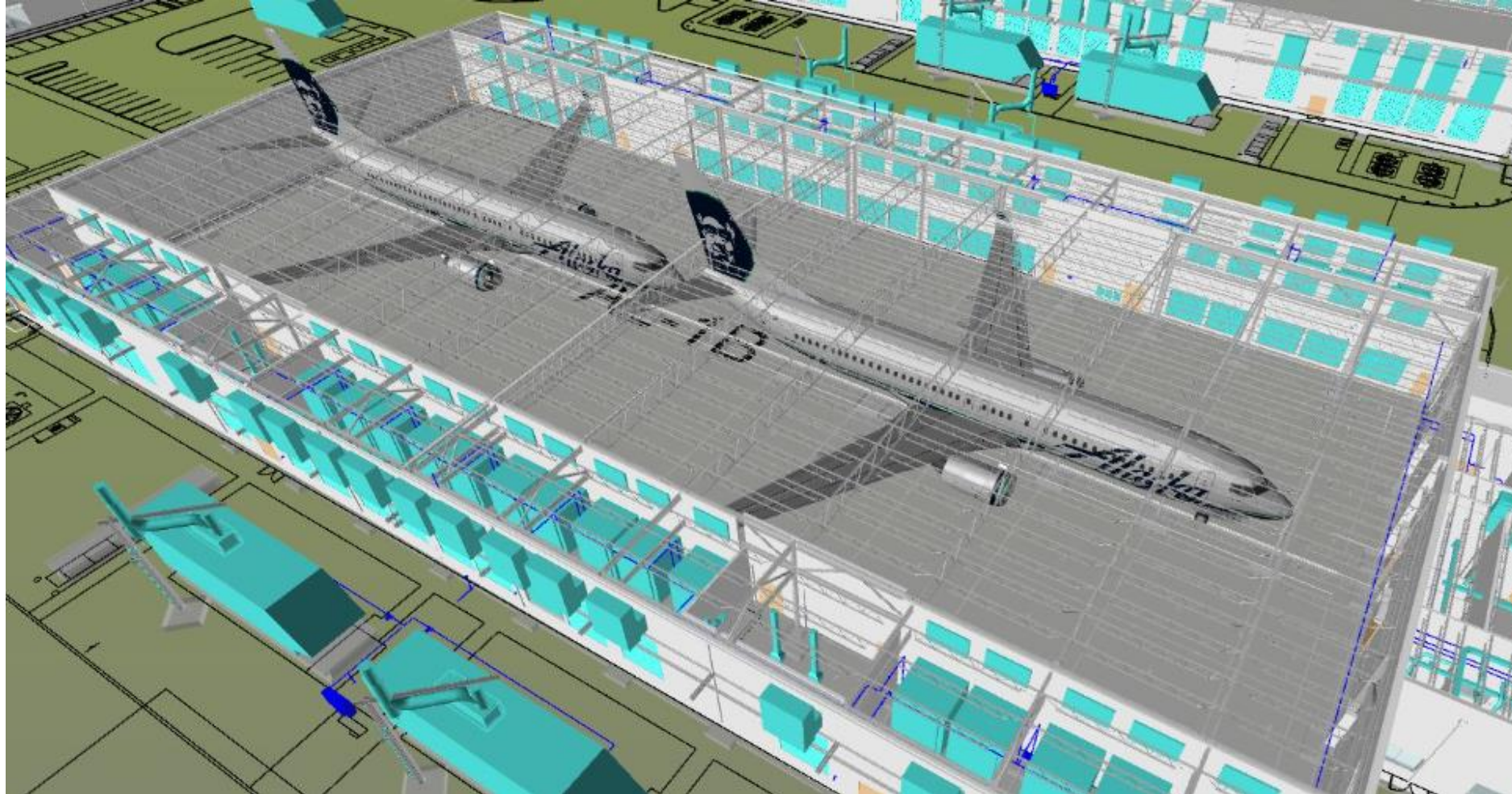
175k+ miles of fiber + subsea cables

185+ Network edge sites

200+ Express route partners

20k+ peering connections

The Cloud is About Building Massive Shared Infrastructure





How much
traffic does our
network carry?

public azure, T0 → servers

15 exabytes (10^{18}) / day

**> 2 TB for every person on earth
every day**

Quality is job #1

Our work here matters to people's real lives

- 911 dispatch systems
- Hospital and medical IT systems
- Doctor's office dictation systems (e.g., Nuance, now part of MSFT)
- The movement of people, goods, and services around the world
- Major retail
- Banking and fintech

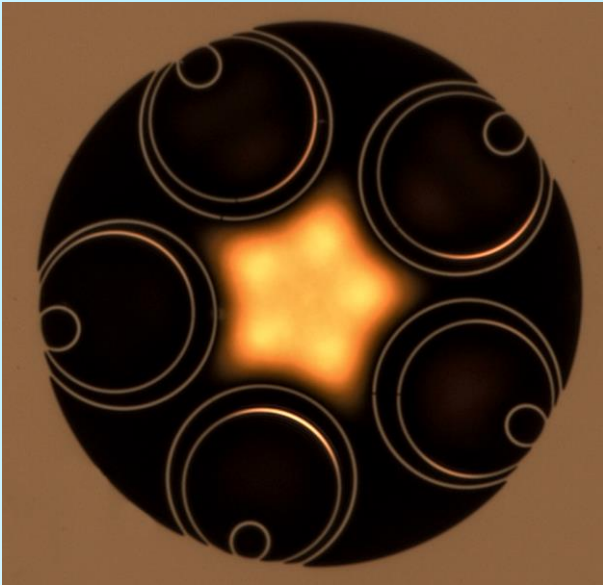
Your neighbors, your families count on you doing quality work

Sources of Disruption: Physical Layer

Hollow Core Fiber

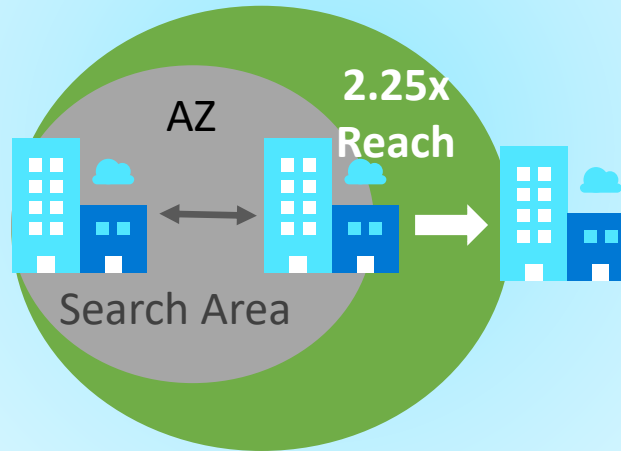
Light travels in gas, not glass

Engineered mirror removes complex physics.



FASTER

47% faster than traditional fiber increases
Region search area by 2.25x.



Physical layers I don't believe in(for data centers)

Wireless

- Key metric: bps/m³
- Signal to interference ratio between two fibers is infinity
 - Even wires are close to infinity

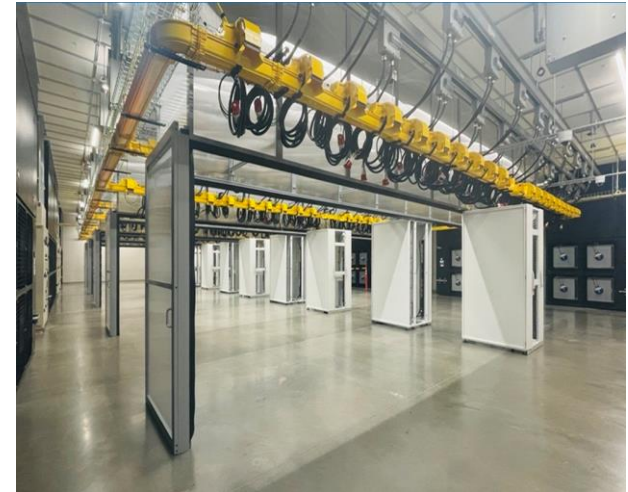
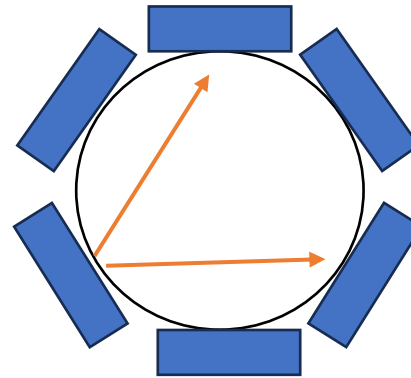
How to keep an antenna from facing 100s of other antennas?



Freespace Optics (lasers)

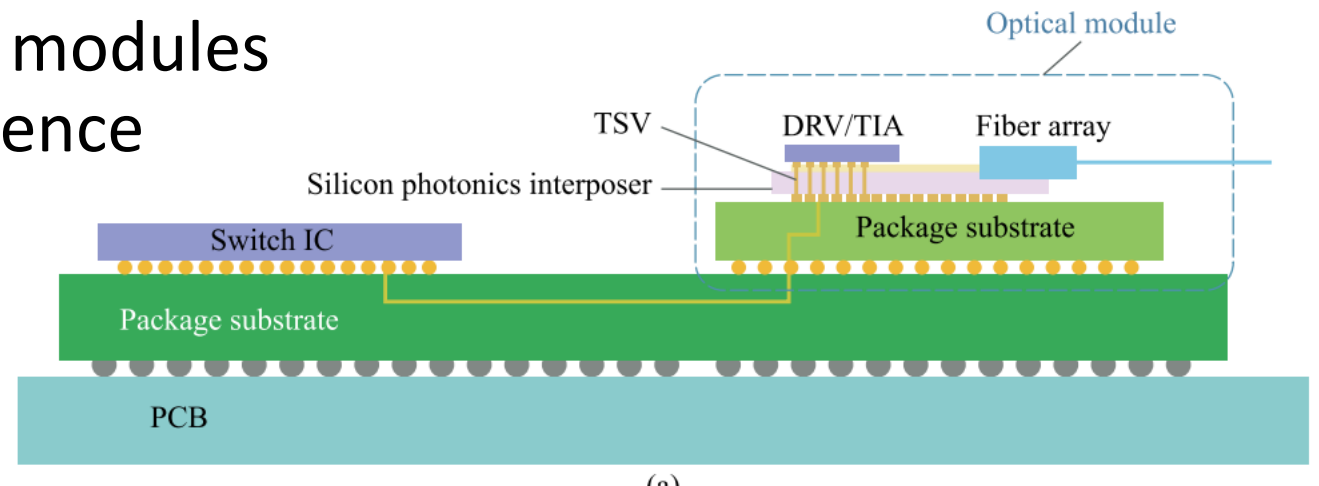
- Waveguides are wonderful
 - Fiber avoids alignment issues
- Incremental build out is important

Packing cylinders or spheres into a world of rectangles is hard



Co-packaged optics

- Bandwidth through pins of package has been a bottleneck for decades
- Electrical Engineers continue to amaze: 100G SerDes once thought impossible – now driving Direct Attach Copper (DAC) cables
- Physical limits exist(?) 200G SerDes?
- Rich ecosystem of multi-chip modules & packages coming into existence



Co-packaged optics

- Small-scale production exists
- Probably mostly engineering to make it large scale
 - Needs multiple vendors, interop, supply chain assurances, QA, yields, ...

Will radically change everything about design of servers, coprocessors, *PUs, racks, networks

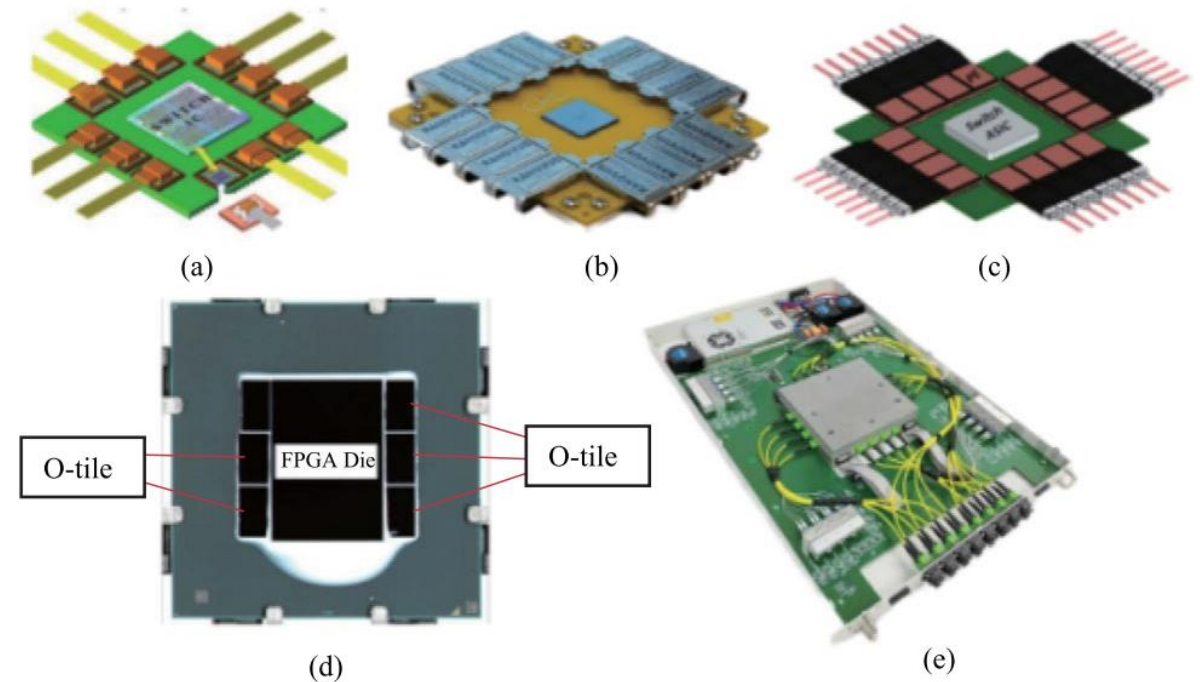


Fig. 18 CPO solutions. **a** IBM and II-VI. **b** Ranovus. **c** Intel. **d** Ayar-labs. **e** Hengtong Rockley

AI/Supercomputing?

Really, it's a networking problem...

Challenge:

Move huge amounts of data among lots of endpoints

Endpoints are specialized coprocessors – GPU, TPU, DPU, XPU, *PU

Those *PUs tend to be HW ASICs

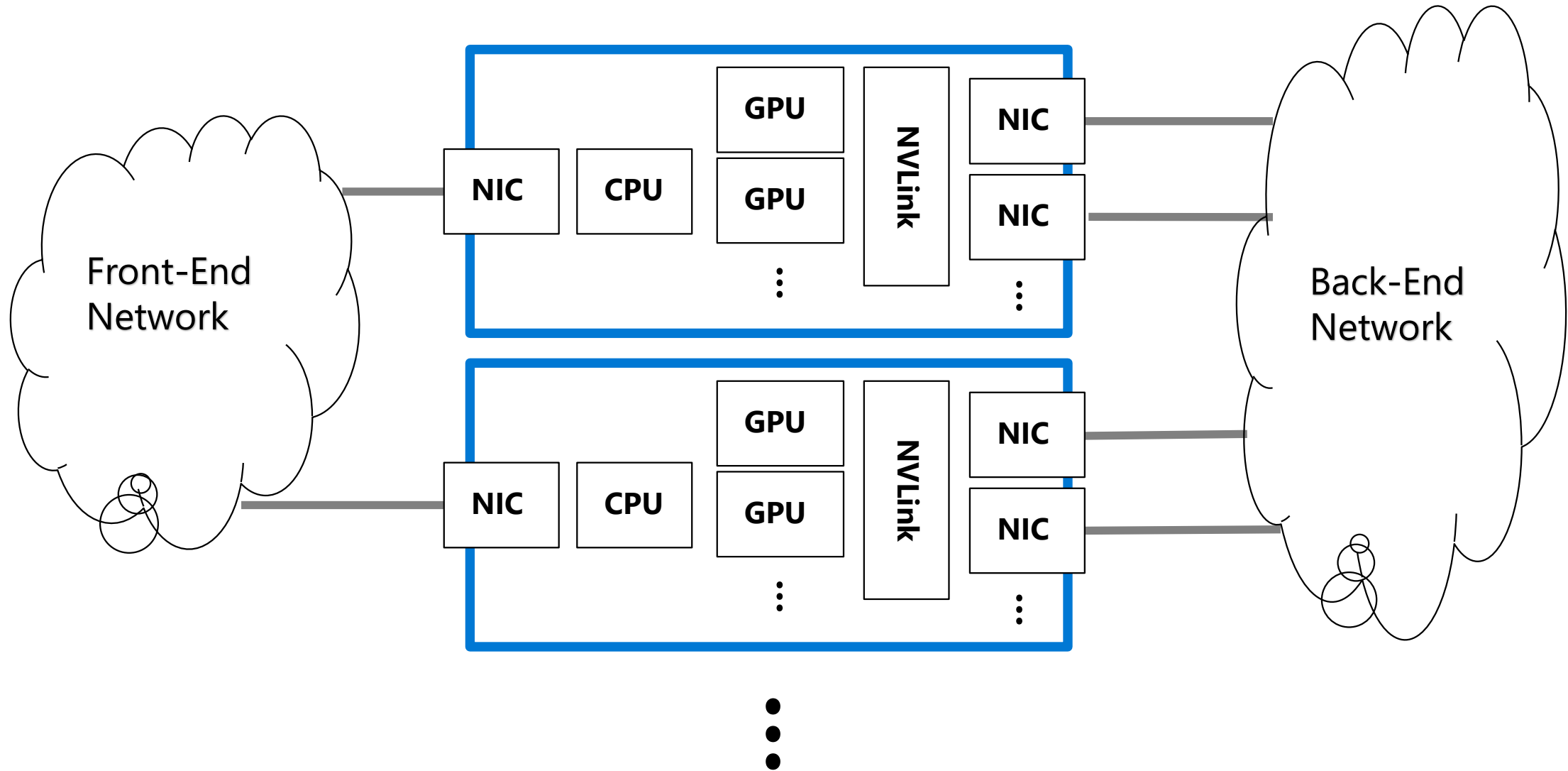
Gaps:

Transport libraries with HW-friendly implementations

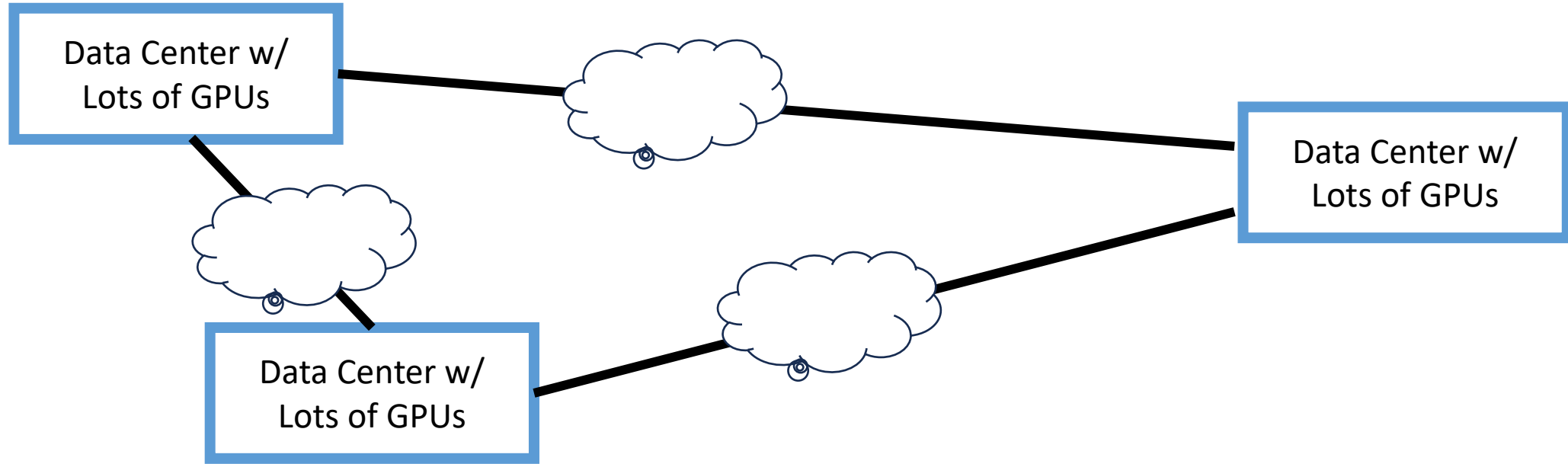
Congestion control (?)

A network layer (?) with virtualization capabilities (?) IP (?)

Typical GPU cluster design



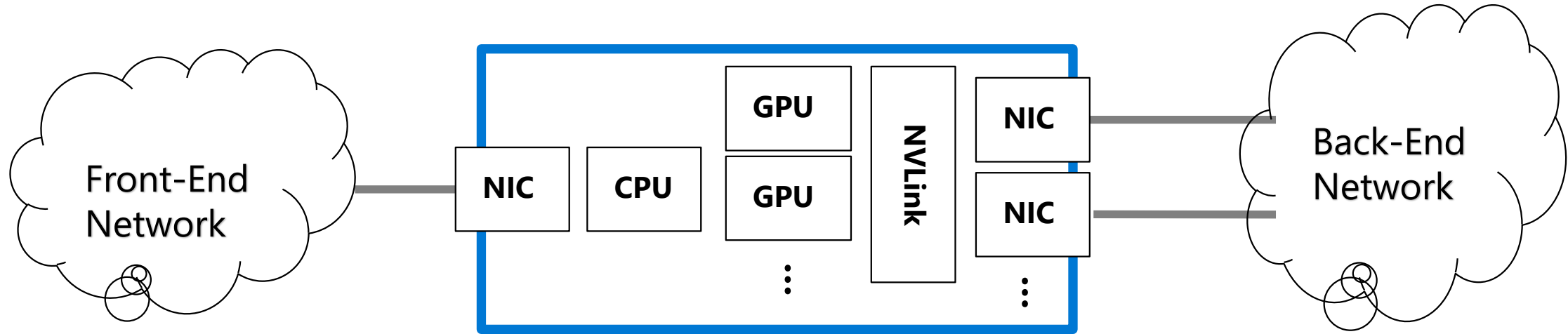
Context: Building AI training clusters is a race



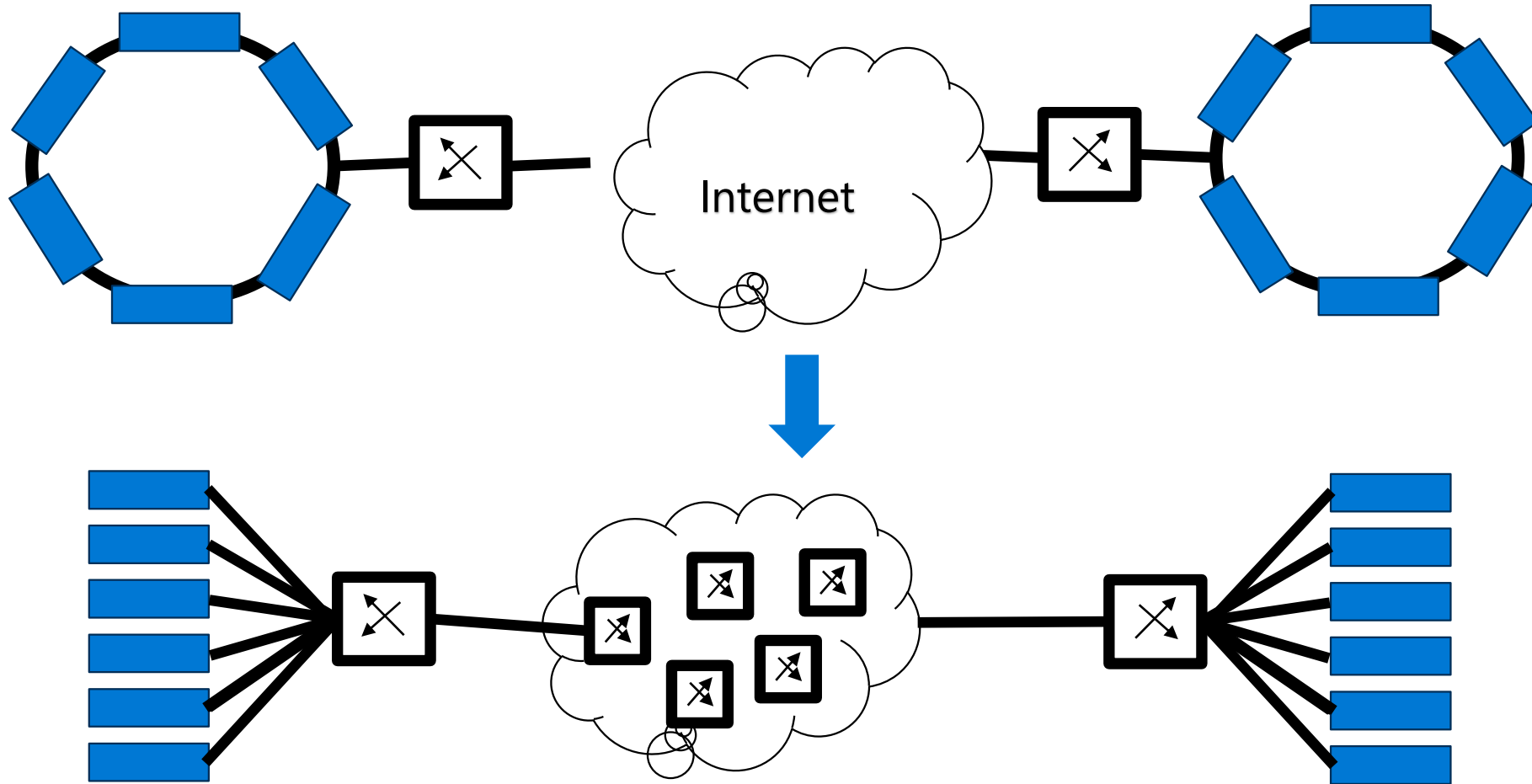
Make the GPUs appear as one physical cluster, despite:

- Different speed of light latencies
- Different bandwidths
- Different link-layer behaviors (flow control, etc)

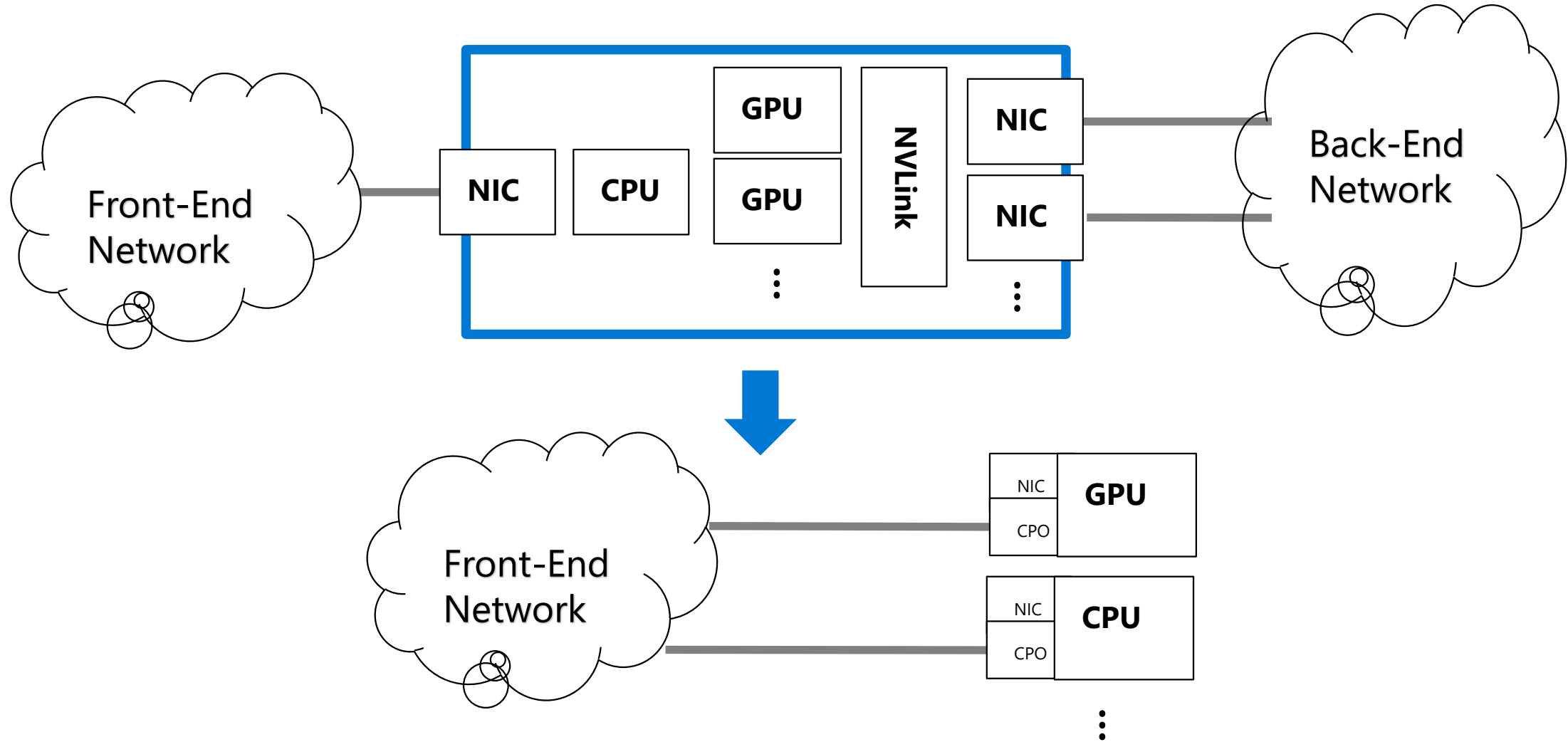
The need for magic specialized layers is short-lived



The need for magic specialized layers is short-lived

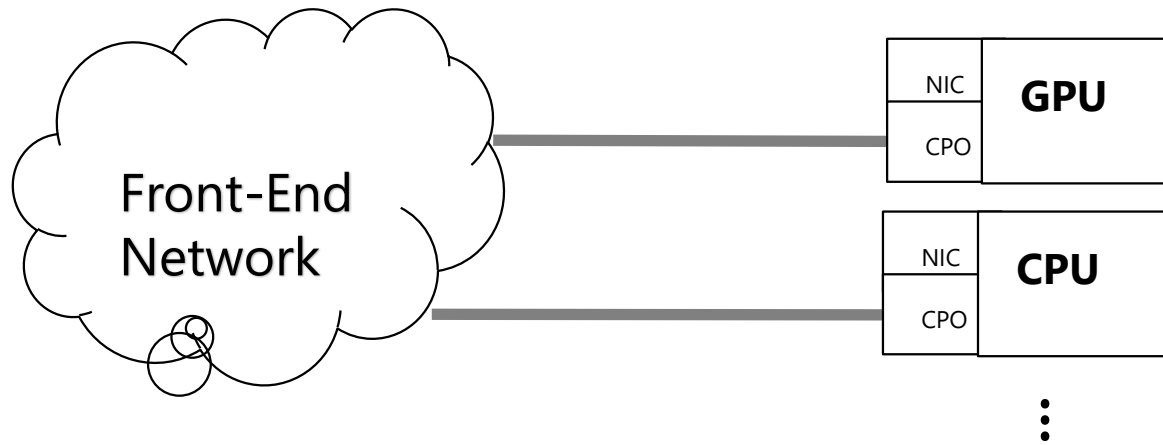


The need for magic specialized layers is short-lived



CPO needed to get data off the package anyway
– make it a complete network

Now there are many research questions...



- Naming?
- Virtualization and multi-tenancy?
- What "verbs" to support?
- Flow control? Error control? In-order delivery?

These protocols become the essence of the disaggregated data center

--- the new computer

--- the new brain? (Depends on your opinions on general AI)

Network Offloads and Middleboxes



- DPDK is great for implementing network middleboxes!
 - \$10K server = very flexible way to process 100 Gbps of traffic
 - Program in C or Rust – many programmers familiar with development process
 - Typically lots of resources left unused (cores, memory)
- Systems optimized for processing network data do much, much better
 - ~ \$10K of DPUs = ~ 1.6 Tbps of network processing

Network Offloads and Middleboxes

DPU Architectures fall into three main categories:

- Sea of (wimpy?) cores
- Cores with a flexible data path (FPGA, P4 engine)
- Cores with fixed accelerators (parser, crypto)

Programming them:

- Run to completion code in C-like language
- OS to handle scheduling lots of threads
- It's all about the memory model – BW, consistency among cores

Representing them in the network:

- Disaggregated APIs for SONiC Hosts

Software Scale Challenges

- **Some Azure statistics**

- > 100M SDN policy objects under management
- > 2.5B requests per day to read, modify or create policy objects
- Each customer expects to create/destroy 20K VMs/min
- Customers want 1M IPs per VNet (mostly due to containers)

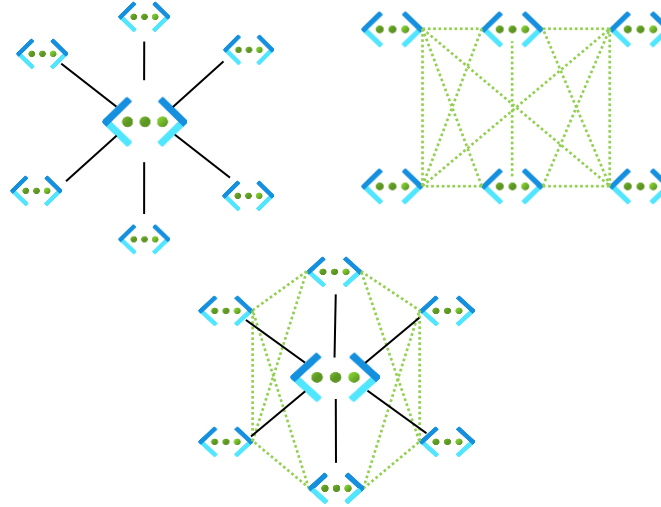
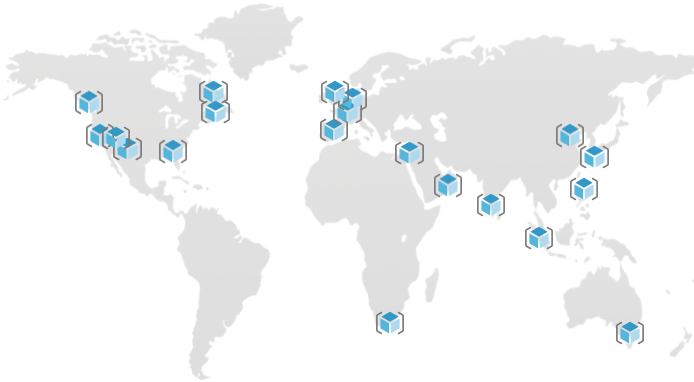
- **Challenge:** Standard trick to get more scale and availability: partition state

- But: customers want seamless update of all their assets
- But: customers want global peering of assets across fault domains
- Approach: Publish/Subscribe systems that do 100M notifications/min over 100M partition keys

- **Challenge:** Goal state must always be correct and consistent

- But: complex multi-operation workflows can fail half-way and code is frequently updated
- Approach: Systems for continual validation and repair of goal state

Policy Scale Challenges



What-if ?

- Imagine company with 1,000s of stores, office, warehouses around the world
- Each site has multiple virtual networks
- Company needs to control security boundaries around different kinds of assets
- Needs different peering arrangements for different virtual networks
- Company needs to safely deploy changes to each region
- Monitoring for health
- Rollback or fix and roll forward

Managing SDN policy now as complicated as managing network devices

Security Challenges

- **Denial of Service Attacks remain challenging**
 - Large L3-4 attacks generally well handled – even in Tbps (<https://azure.microsoft.com/en-us/blog/azure-ddos-protection-2021-q3-and-q4-ddos-attack-trends>)
 - L7 attacks becoming more common in 2023 and harder to defend
 - LMMs to the rescue for generating attack signatures?
 - Application and protocol improvements to make throttling and scale up/down easier?
- **How do we create the “pit of success” for developing secure apps?**
 - We need an end-to-end view of developing applications that are secure
 - Better sandboxes and taller walls are not enough
 - Certs are hard: rotating them, verifying them, ...
 - Building extensible or updatable services is hard: Log4j, malware in OSS repositories

Quantum Networking



